

LATNIVAARA, Sweden

WMO: 020380

Lat: 67.2551N

Lon: 20.2704E

Elev: 1486

StdP: 13.92

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-19.8	-31.0	1.0	-24.5	-25.4	1.4	-19.4	19.4	27.9	16.7	24.4	0.7	340	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	73.8	59.0	70.2	57.6	66.7	56.1	62.0	70.4	59.7	67.0	57.6	63.8	6.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	78.1	64.1	56.6	72.1	62.3	54.6	67.0	60.1	28.5	70.6	26.9	67.2	25.4	63.9	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	14.3	12.3	-30.9	78.2	5.3	4.5	-34.7	81.4	-37.8	84.1	-40.8	86.6	-44.7	89.9
			-30.9	64.0	5.2	2.4	-34.7	65.7	-37.8	67.1	-40.7	68.4	-44.5	70.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	6.4	7.8	16.4	27.9	39.0	48.4	54.4	50.1	41.6	28.5	16.9	12.0
	DBStd	19.00	13.22	13.18	10.18	7.96	7.01	6.60	5.56	5.96	6.31	8.67	10.93	12.41
	HDD50	7884	1351	1182	1041	664	350	107	20	73	259	666	993	1178
	HDD65	13057	1816	1602	1506	1114	807	499	331	463	702	1131	1443	1643
	CDD50	306	0	0	0	0	8	58	157	75	8	0	0	0
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	96	0	0	0	0	0	21	66	9	0	0	0	0
	CDH80	7	0	0	0	0	0	2	5	0	0	0	0	0
Wind	WSAvg	5.1	3.7	4.2	5.8	5.8	6.3	6.4	5.7	5.0	5.4	4.6	3.8	4.2
Precipitation	PrecAvg	21.8	1.2	1.0	0.9	0.9	1.8	2.3	3.9	2.8	2.0	1.8	1.6	1.3
	PrecMax	26.9	2.3	1.9	1.9	1.8	3.4	4.9	6.2	4.8	4.3	3.5	3.3	2.4
	PrecMin	15.8	0.1	0.2	0.4	0.1	0.3	0.6	1.7	1.0	0.9	0.8	0.4	0.3
	PrecStd	3.7	0.6	0.5	0.4	0.5	0.9	1.0	1.3	1.1	1.0	0.7	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	40.3	42.6	54.3	71.5	77.1	80.0	75.1	63.4	50.6	41.9	39.2
		MCWB	36.0	36.3	36.9	43.5	52.4	60.9	63.1	61.7	53.9	45.9	38.6	36.0
	2%	DB	34.4	35.3	38.3	47.7	64.8	71.0	75.3	70.9	59.1	46.9	37.7	34.4
		MCWB	32.2	32.6	33.9	39.5	51.0	54.2	60.4	59.6	51.7	43.6	36.2	32.1
	5%	DB	29.4	30.9	35.3	43.9	58.4	66.8	71.7	66.9	55.7	44.1	33.7	31.1
		MCWB	28.3	29.2	31.8	37.3	47.9	53.3	59.3	58.0	50.2	41.5	32.9	30.2
	10%	DB	24.9	27.3	32.3	40.8	52.7	62.6	67.7	63.0	52.9	40.9	31.5	28.9
		MCWB	24.1	26.2	29.5	35.5	44.0	51.9	57.4	55.5	48.8	39.0	31.0	28.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	36.8	36.8	44.0	54.9	62.1	65.3	65.0	55.8	47.1	39.4	36.2
		MCDB	39.2	40.2	41.8	53.7	67.1	74.3	74.2	72.3	59.8	48.7	41.0	38.8
	2%	WB	32.4	32.6	34.3	40.5	51.8	57.4	62.9	61.0	53.2	44.1	36.2	32.4
		MCDB	34.3	35.0	37.8	46.7	63.7	64.3	72.0	67.7	57.6	46.4	37.9	33.7
	5%	WB	28.4	29.3	32.1	38.0	48.1	54.9	60.8	59.0	51.3	41.5	33.0	30.3
		MCDB	29.4	31.0	34.9	43.0	58.0	63.3	68.4	65.7	54.3	43.7	33.7	31.1
	10%	WB	24.2	26.2	29.7	35.9	45.0	52.7	58.7	56.6	49.6	39.1	31.0	28.0
		MCDB	24.9	27.3	32.1	40.5	51.6	61.0	65.6	61.9	52.2	40.7	31.4	28.9
Mean Daily Temperature Range	5% DB	MDBR	16.8	18.5	20.1	18.2	17.4	19.0	19.9	17.2	14.2	14.1	15.8	
		MCDBR	15.5	15.9	19.3	20.8	29.0	27.7	27.9	27.5	22.1	16.6	11.3	14.1
		MCWBR	14.6	14.4	15.9	13.5	17.1	14.1	14.9	16.8	15.0	13.2	10.6	13.5
	5% WB	MCDBR	15.2	15.4	18.2	18.8	26.9	22.4	22.7	24.3	17.7	15.1	11.5	13.8
		MCWBR	14.5	14.1	15.3	12.3	16.2	12.4	13.5	15.4	12.6	12.2	10.9	13.4
Clear-Sky Solar Irradiance	taub		0.173	0.236	0.272	0.301	0.320	0.324	0.332	0.318	0.288	0.260	N/A	N/A
	taud		1.934	2.148	2.234	2.277	2.463	2.519	2.508	2.529	2.586	2.383	N/A	N/A
	Ebn at Noon		60	193	248	272	279	281	274	267	251	188	N/A	0
	Edh at Noon		3	15	24	31	29	28	28	24	18	12	N/A	0
All-Sky Solar Radiation	RadAvg		17	173	638	1234	1558	1602	1496	1093	602	207	38	0
	RadStd		1	17	35	58	139	127	149	96	78	32	4	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page